JUST Student Assistant

Jordan University of Science and Technology (JUST)

*An LLM-Powered IT Support Assistant for JUST Students

Phase 1: Problem and Data

1. Team and Roles:

Team Members	ID	Role
Rand Al-Darabseh	174269	Product Lead
Batool Abualhayjaa	168361	Data Lead
Hiba Al-Naimat	170251	Evaluation Lead & Engineering Lead

2. User and Problem

Target Users:

The target users of the JUST Student Assistant are students at Jordan University of Science and Technology (JUST) who need reliable and accessible information about university services, academic resources, and student-related matters.

Problem:

Students often need information about different university services, including IT support, student affairs, health services, academic resources, counseling, and administrative procedures. However, finding accurate and up-to-date information across different university departments and websites can be time-consuming and sometimes confusing.

Students may struggle to identify the correct department, understand service procedures, find contact information, or determine the requirements for specific services. This creates a need for a centralized and accessible assistant that helps students find relevant university information efficiently.

Current Solution:

Currently, students rely on university websites, official documents, department offices, online searches, classmates, and direct communication with university staff to obtain information. However, information may be distributed across multiple sources, making it difficult to find the correct answer quickly. Some procedures and service details may also change over time, requiring students to verify information with the relevant university departments.

Evidence of the Problem:

A survey was conducted with 41 respondents, including 39 JUST students. The results showed that 58.5% experienced Wi-Fi or internet issues, 51.2% faced e-learning platform problems, and 70.7% reported that technical issues affected their studies or assignments. Additionally, 82.9% expressed willingness to use an AI-powered IT support assistant.

These findings demonstrate the need for accessible and reliable digital support for students, particularly regarding university IT services. The project builds on this initial evidence and expands its scope to cover additional university services and student-related information.

3. What the System Will Do

The JUST Student Assistant will help students at Jordan University of Science and Technology (JUST) access reliable information about university services, academic resources, and student-related inquiries.

The system will use Large Language Models (LLMs), Continued Pre-Training (CPT), Supervised Fine-Tuning (SFT), and Retrieval-Augmented Generation (RAG) to improve its ability to understand student questions and provide relevant, clear, and reliable answers.

Students will be able to ask questions in natural language, in Arabic or English, about different university services, including IT support, student affairs, health services, academic resources, counseling, and administrative procedures.

The system will use information collected from official JUST documents, university websites, service guidelines, and other trusted university resources to support its responses.

The assistant aims to provide organized answers, explain service procedures, share relevant contact information, and guide students to the appropriate university departments when necessary.

If the required information is unavailable or cannot be verified from the available sources, the assistant will inform the student rather than generate unsupported information. Information that may change over time, such as fees, working hours, and eligibility requirements, should be verified through the relevant university department.

The following examples illustrate possible interactions between the user and the system:

User Input	**System Answer**
How can I connect to the university Wi-Fi?	The assistant provides the available university Wi-Fi connection instructions and explains the required steps.
What services does the JUST Health Center provide?	The assistant retrieves information about the available medical clinics and healthcare services at the university health center.
How can I book an appointment at the Smoking Cessation Clinic?	The assistant provides the available appointment booking procedures and relevant contact information.

What student financial aid services are available at JUST?	The assistant explains the available student loan, scholarship, and employment services based on official university information.
What support is available for students with disabilities?	The assistant provides information about the Access and Inclusive Education Platform and the available academic and administrative support.
What should I do if I cannot access JUST Learn?	The assistant retrieves the relevant troubleshooting instructions and provides the available steps to resolve the login issue.

4. Data

The project will use official IT-related documents and resources published by Jordan University of Science and Technology (JUST), including student IT support instructions, user guides, troubleshooting procedures, and information about university IT services such as student accounts, university email, Wi-Fi connectivity, and the e-learning platform.

The main data sources will include official university websites, departments, and service portals that provide reliable information relevant to JUST students.

The team will collect and review relevant documents from these sources to ensure the information is accurate, reliable, and suitable for answering common student inquiries.

Data Sources:

1. **JUST IT & Communications Center (ITCC)**
2. **JUST Learn – Center for E-Learning and Open Educational Resources**
3. **JUST Student Services Portal**
4. **JUST Department of Student Services**
5. **JUST Health Center**
6. **JUST Student Life**
7. **Other Official JUST University Resources**

5. Planned Architecture

The system will use Retrieval-Augmented Generation (RAG) to retrieve relevant information from trusted and official JUST documents and resources. An LLM will then generate clear and helpful answers based on the retrieved information, covering different student needs such as IT support, student affairs, health services, academic resources, counseling, and university procedures. The system will provide source references to help students verify the information.

6. How We Will Measure Success

Success will be measured by:

- **Answer Accuracy:** Providing correct and relevant answers to student questions.
- **Source Reliability:** Using trusted and official JUST sources.
- **Hallucination Prevention:** Reducing unsupported or incorrect information.
- **Information Coverage:** Addressing questions across different university services.
- **Answer Clarity:** Providing clear and understandable responses in Arabic and English.

7. Risks

Risk	Mitigation
Missing or incomplete documents	Find alternative official JUST sources.
Incorrect or outdated answers	Use reliable sources and provide references.
Limited time	Prioritize the most important student services.
Inconsistent information across sources	Prefer official and updated university sources.
Questions outside the system's scope	Inform students when the requested information is unavailable.